Digital Twin-Based Autonomous Navigation and Control of Omnidirectional Mobile Robots

Hongjiu Yang, Ziqi Qin, Yuanqing Xia, *Fellow, IEEE*, and Fuyang Cheng

***Abstract*—In this paper, an autonomous navigation and control method is proposed for an omnidirectional mobile robot (OMR) based on digital twin (DT). An OMR-based DT system is designed from four dimensions, i.e., physical space, virtual space, application service and data processing. By the DT system, the autonomous navigation and control method is implemented from two steps: Autonomous navigation in virtual space and physical-virtual motion synchronization. The autonomous navigation of a virtual OMR is completed by fusion of an Astar algorithm and a dynamic windows approach in virtual environment. After obtaining an optimal navigation trajectory in virtual space, a physical OMR is driven by a physical-virtual motion synchronization method. Integral sliding mode controllers are designed to enhance robustness of the physical-virtual motion synchronization such that velocity tracking errors between the physical OMR and the virtual OMR are convergent. Experiment results show effectiveness of the DT-based autonomous navigation and control method by a physical-virtual synchronization tracking error 0.061 m and capability to handle various tasks.**

***Index Terms*—Digital twin (DT), omnidirectional mobile robots (OMRs), autonomous navigation, dynamic windows approaches (DWAs), integral sliding mode control.**

I. Introduction

ROBOT autonomous navigation and control technology, which enables mobile robots to plan collision free trajectories and move autonomously to destination, is widely used in fields such as industrial automation [1], logistics [2] and warehousing [3]. There exist high algorithm development and validation costs, complex algorithm deployment and high challenge in collaborating with intelligent agents for traditional schemes on autonomous navigation and control technology [4]. Digital twin (DT) technology can offer a mapping relationship between physical space and virtual space, which proposes a new idea on autonomous navigation and control for mobile robots [5].

DT technology is to create virtual space that is completely equivalent to physical space in a computer [6], [7], [8]. The virtual space flexibly reflects three-dimensional operating states of the physical space, which is conducive to equipment monitoring and rapid fault localization [9], [10], [11]. With powerful computing power of the virtual space, complex decisions and processes can be simulated, predicted and optimized before being applied to the physical space [12], [13]. In [14], DT greatly improves efficiency and autonomy of robot measurement systems by providing core tasks such as collision detection, sensor simulation and path planning. Advantages of DT inspire us to apply DT technology in autonomous navigation and control for mobile robots to improve efficiency and reduce costs.

By DT technology, autonomous navigation and control algorithms can be developed and optimized in virtual space. As a core issue of autonomous navigation, path planning is responsible for determining a collision free trajectory from start point to end point [15]. Common robot path planning algorithms include Dijkstra [16], Astar algorithms [17], Artificial Potential Field [18], Ant Colony Algorithms [19], etc. As a heuristic search algorithm, Astar algorithms have advantages in finding the shortest global path in complex graphics in a short period of time [20]. However, the global path planned by Astar algorithms is a series of coordinate points that is not suitable for directly driving the robot in virtual space. Dynamic window approaches (DWAs) are path planning algorithms specifically designed for robots, which adjusts robot velocity in real-time to avoid obstacles and reach an endpoint [21]. In [22], a long-term DWA is applied to generate a global trajectory in both static and crowd environments. In [23], a DWA is used for high-speed obstacle avoidance in unknown environments. Obstacle avoidance effectiveness of an improved DWA is confirmed in urban road environments in [24]. However, traditional DWAs are easy to fall into local optima since only one endpoint are used for DWAs as a guide [25]. It inspires us to choose a point in a global path planned by Astar algorithms as a guide to run DWAs, which balances local and global planning pathes. Therefore, it is reasonable to fuse Astar algorithms and DWAs for autonomous navigation and control of a robot in virtual space.

DT migrates the complex autonomous navigation and control problems of a physical space to a virtual space with strong computing power and much flexible debugging. The next challenge is to ensure physical-virtual motion synchronization, so that a physical robot reproduces the optimal trajectory calculated by a virtual robot in physical space. If the virtual space is a perfect mirror of the physical space, then the synchronization will be completed by controlling the physical robot with the same control signal as the virtual robot [26], [27], [28].

Received 4 January 2024; revised 15 March 2024, 9 May 2024, 17 July 2024, and 21 October 2024; accepted 17 December 2024. Date of publication 23 December 2024; date of current version 18 April 2025. This work was supported by the National Natural Science Foundation of China under Grant 62373269 and Grant 62333016. The review of this article was coordinated by Prof. Zhaolong Ning. *(Corresponding author: Hongjiu Yang.)*

Hongjiu Yang, Ziqi Qin, and Fuyang Cheng are with the Tianjin Key Laboratory of Intelligent Unmanned Swarm Technology and System, School of Electrical and Information Engineering, Tianjin University, Tianjin 300072, China (e-mail: yanghongjiu@tju.edu.cn; qinziqi1143@tju.edu.cn; fuyangcheng@tju.edu.cn).

Yuanqing Xia is with the School of Automation, Beijing Institute of Technology, Beijing 100081, China (e-mail: xia_yuanqing@bit.edu.cn).

Digital Object Identifier 10.1109/TVT.2024.3520991

0018-9545 © 2024 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Universidad Federal de Pernambuco. Downloaded on July 31,2026 at 16:24:55 UTC from IEEE Xplore. Restrictions apply.

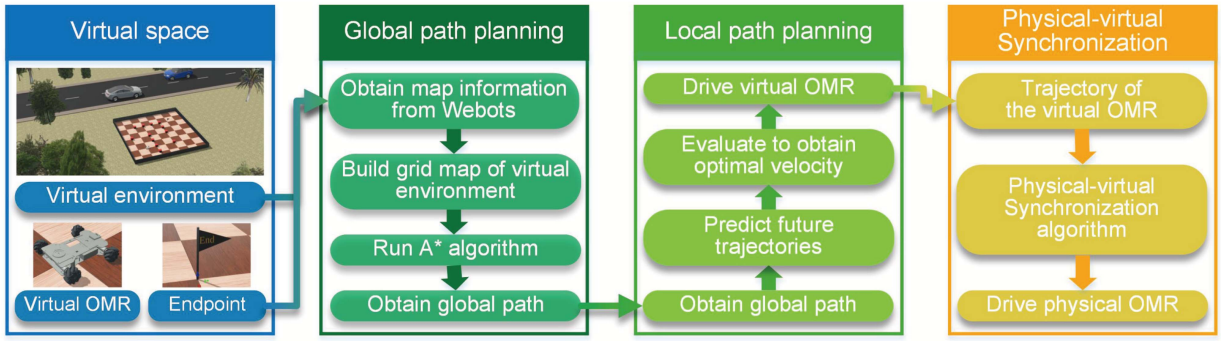


Fig. 1. Process of the DT-based autonomous navigation method.

Unfortunately, it is impossible to accurately model unknown disturbances in physical environment and complex dynamics of the physical robot [29]. Actually, motion synchronization problems of multi-agent under disturbance are widely solved by control methods in physical space [30], [31]. In [30], a synchronization control strategy based on position errors is proposed for a grid connected multi axis motor system to solve impact of compound disturbances on synchronization performance. Integral sliding mode control has advantages of strong robustness, fast rate of convergence and high control precision [32], [33]. In [32], an integral sliding mode control strategy is proposed for motion tracking control of a piezoelectric-driven nanopositioning system. Therefore, it is feasible to utilize integral sliding mode control to ensure accuracy and robustness of physical-virtual motion synchronization.

In this paper, an autonomous navigation and control method is designed for an omnidirectional mobile robot (OMR) based a DT system. It is important to enhance intelligence and effectiveness of the autonomous navigation and control method for achieving precise mapping of motion relationships between the virtual OMR and the physical OMR while generating globally optimal trajectories. The main contributions in this paper are clarified in the following.

1) An autonomous navigation method is arranged in a DT system to improve trajectory solving accuracy and sure calculation speed.
2) An Astar algorithm and a DWA are integrated to solve a trajectory iteratively and prevent the DWA from converging to local optima.
3) A unified control structure is designed to achieve motion synchronization and torque control for the physical OMR and the virtual OMR.

The remaining of this paper is organized as follows. In Section II, a DT system is established to arrange algorithms and improve solving accuracy of trajectories. In Section III, kinematic and dynamic models of the physical OMR and the virtual OMR are designed. Based on the DT system, an autonomous navigation and control method is proposed for the physical OMR in Section IV. The verification of the method proposed in this paper is shown in Section V by real-world experiments. Finally, the main conclusions are presented in Section VI.

Notations: In this paper, the transpose of a matrix P is expressed as P^{T}, $|\cdot|$ is used to denote an absolute value, $\mathrm{sign}(\cdot)$ donates a standard sign function, $A[j]$ represents the element of a vector $A \in \mathbb{R}^{n\times 1}$ corresponding to the index j, $1 \le j \le n$.

II. Structure of the OMR-Based DT System

A. Problem Statement

In this paper, an autonomous navigation and control method based on DT is proposed for the physical OMR. A process of the autonomous navigation and control method is illustrated in Fig. 1. In Fig. 1, a manager firstly sets an endpoint in the high-fidelity virtual environment by the application services. The virtual endpoint corresponds to the expected endpoint of the physical OMR in the physical environment. A grid map is built by automatically obtaining virtual map information including environmental boundaries, obstacles coordinates and sizes, robot coordinate and size, and endpoint coordinate. A global path from the virtual OMR to the endpoint is planned by an A-star algorithm in grid map. DWA calculates optimal collision free velocity to drive the virtual OMR in real-time. Debugging and optimizing algorithms by simulation of the virtual OMR while the physical OMR is offline. The physical OMR is connected to the DT system and synchronized with the virtual OMR for navigation movement based on the physical-virtual synchronization algorithm.

B. The DT System

Establishment of a DT system is a primary task for application of the DT technology in autonomous navigation and control of an OMR. In this paper, a DT system is designed from four dimensions: Physical space, virtual space, application service and data processing. Composition and connection of each dimension are illustrated in Fig. 2. Meanwhile, the project code of this article is uploaded to Github as *https://github.com/FY-Cheng/digital-twin*.

The physical space is composed of a physical OMR with four Mecanum wheels and its operating environment shown as the orange part in Fig. 2. As ultimate controlled object of the DT system, the physical OMR is mainly responsible for transmitting perceptual information about oneself and environment and responding received control command. Therefore, a Jetson Xavier NX processor with a robot operating system (ROS) and a STM32 microcontrol unit are equipped in the

Authorized licensed use limited to: Universidad Federal de Pernambuco. Downloaded on July 31,2026 at 16:24:55 UTC from IEEE Xplore. Restrictions apply.

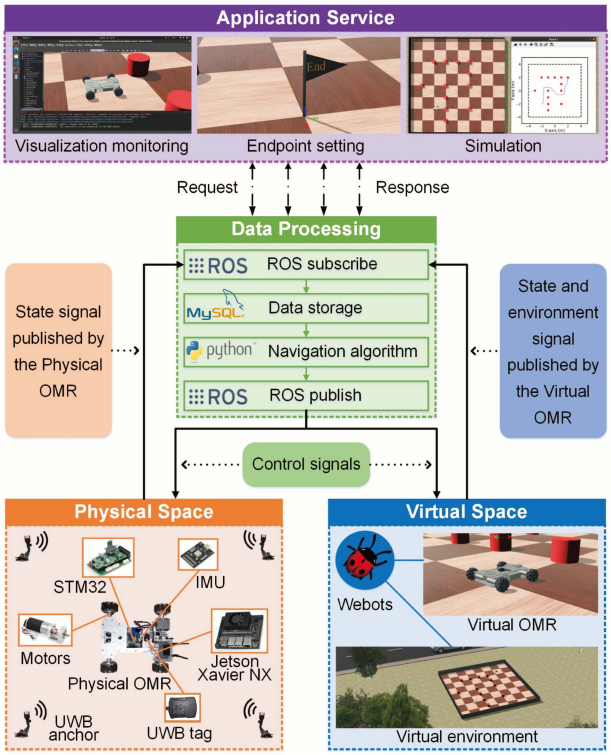


Fig. 2. Four-dimensional structure of the OMR-based DT system.

physical OMR for communication and motors control. Velocity information of the physical OMR is calculated through encoder signals of four wheel motors. Furthermore, four ultra wide band (UWB) anchors are place at four corners of a square to define a physical world coordinate system. An UWB tag is equipped in the physical OMR for acquisition of position information. Yaw and yaw rate of the physical OMR during operation is determined by an inertial measurement unit (IMU) measuring submodule.

The virtual space is a high fidelity mirror model of the physical space in geometric, mathematical and physical dimensions. As shown in blue box of Fig. 2, the virtual environment and the virtual OMR are built by Webots with the same geometric structure as the physical space. The same world coordinate system is set up for the virtual environment in Webots. Motors and encoders are also equipped in four wheels for motion control of the virtual OMR. Meanwhile, GPS and IMU are installed in the virtual OMR to obtain accurate position and angular information. By collaboration with Webots, ROS is utilized for manipulating the virtual OMR.

In purple area of Fig. 2, the application service is responsible for providing two main services to manager of the OMR-based DT system: 3-D real-time visualization monitoring of the physical OMR and navigation endpoint online setting. The 3-D real-time visualization monitoring allows manager to keep abreast of real-time motion status of the physical OMR by viewing the virtual OMR of Webots interface. High-fidelity virtual space modeling and a physical-virtual synchronization algorithm ensure accuracy and credibility of the 3-D real-time visualization monitoring. The navigation endpoint online setting optimizes

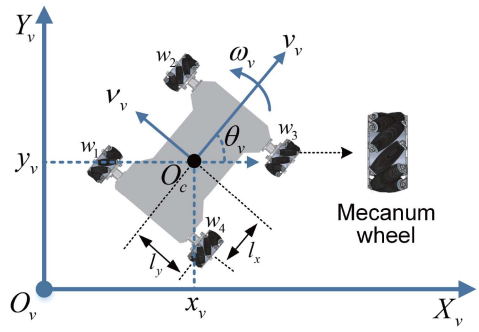


Fig. 3. Kinematic analysis of the virtual OMR.

the autonomous navigation of physical OMR by the following steps: 1) Find the point corresponding to navigation endpoint of the physical OMR in the virtual environment. 2) Take the physical OMR offline and implement autonomous navigation and control of the virtual OMR by designing global and local path planning algorithms in the virtual space. 3) Take the physical OMR online after debugging the algorithms in the virtual space, the virtual OMR and the physical OMR navigate autonomously to the endpoint due to the physical-virtual synchronization.

The data processing takes responsibility to deal with twin data from the physical space, the virtual space and the application service as illustrated in green part of Fig. 2. Python, MySQL database and ROS are employed for construction of the data processing. On the one hand, the data processing receives state data from the physical OMR and the virtual OMR by ROS and navigation requirements from the application service for storage and analysis. On the other hand, control signals are calculated by navigation and control algorithms programmed in the data processing and send to drive the physical OMR and the virtual OMR by ROS. Processing and transmission of twin data makes sure normal operation of the OMR-based DT system.

Remark 1: The proposed method in this paper is aimed to apply in all static obstacle avoidance scenarios, for instance, unmanned warehousing environments characterized by numerous static shelves and mobile robots. A virtual OMR is established within the Webots simulation software depending on precise physical data. The DT system, constructed with different software such as Webots, ROS, Python, and MySQL, is established to connect the virtual OMR and the physical OMR. Based on the DT system, the autonomous navigation algorithm is deployed on the server to enhance trajectory solving accuracy.

III. System Modelling and Preliminaries

A. Kinematic Model

In this section, analysis is carried on kinematic and dynamic models of the physical OMR and the virtual OMR. Since the virtual OMR is a digital mirror of the physical OMR, the virtual OMR is analyzed as an example in Fig. 3. In Fig. 3, $\{X_v, O_v, Y_v\}$ is the virtual world coordinate system, O_c is the geometric center of the virtual OMR, (x_v, y_v) is coordinate of O_c in $\{X_v, O_v, Y_v\}$, θ_v is the angle between X_v-axis and forward direction of the virtual OMR, v_v, ν_v

Authorized licensed use limited to: Universidad Federal de Pernambuco. Downloaded on July 31,2026 at 16:24:55 UTC from IEEE Xplore. Restrictions apply.

and ω_v represent the forward velocity, the lateral velocity and the angular velocity of the virtual OMR, respectively. Let $Q_v(t) = [x_v(t), y_v(t), \theta_v(t)]^{\mathrm{T}}$ represent the position and angle information vector. Let $V_v(t) = [v_v(t), \nu_v(t), \omega_v(t)]^{\mathrm{T}}$ be the body velocity vector. Four Mecanum wheels are marked as (w_1, w_2, w_3, w_4) in Fig. 3, the radius of each wheel is R, l_x and l_y are the wheelbase and wheel spacing, respectively. In [9], a kinematic model of the virtual OMR is obtained as

$$\dot{Q}_v(t) = T(\theta_v(t))V_v(t) \tag{1}$$

where $T(\eta)$ is a nonsingular function matrix as

$$T(\eta) = \begin{bmatrix} \cos(\eta) & -\sin(\eta) & 0 \\ \sin(\eta) & \cos(\eta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{2}$$

By the same way, a kinematic model of the physical OMR is expressed as

$$\dot{Q}_p(t) = T(\theta_p(t))V_p(t) \tag{3}$$

where $Q_v(t) = [x_p(t), y_p(t), \theta_p(t)]^{\mathrm{T}}$ is the position and angle information vector, $V_p(t) = [v_p(t), \nu_p(t), \omega_p(t)]^{\mathrm{T}}$ is the body velocity vector.

B. Dynamic Model

In [34], a dynamic model of the virtual OMR is established by employing a Lagrange method. Then a Lagrange function is obtained as

$$L(t) = E_K(t) - E_P(t) \tag{4}$$

where $E_K(t)$ and $E_P(t)$ denote the kinetic energy and the potential energy of the virtual OMR, respectively. In this paper, $E_P(t) = 0$ holds since the virtual OMR runs on a horizontal plane. Based on the Kinetic energy theorem and Fig. 3, $E_K(t)$ is calculated as

$$E_K(t) = \frac{1}{2}m\left(v_v^2(t) + \nu_v^2(t)\right) + \frac{1}{2}I_z\omega_v^2(t) + \frac{1}{2}I_w\sum_{i=1}^{4}\varphi_i^2(t) \tag{5}$$

where m is the total mass, I_z and I_w represent the body moment of inertia and the wheel moment of inertia, respectively. Note that $\Omega_v(t) = [\varphi_1(t), \varphi_2(t), \varphi_3(t), \varphi_4(t)]^{\mathrm{T}}$ represents the angular velocity vector of the four wheels. According to the equations (4) and (5), the dynamic model of the virtual OMR is calculated as

$$\frac{d}{dt}\frac{\partial L(t)}{\partial \dot{Q}_v(t)} - \frac{\partial L(t)}{\partial Q_v(t)} + \frac{\partial \Upsilon(t)}{\partial \dot{Q}_v(t)} = \Gamma_v(t) + \Gamma_{dv}(t) \tag{6}$$

where $\Gamma_v(t) \in \mathbb{R}^{4\times 1}$ represents the input torque vector of four Mecanum wheels, $\Gamma_{dv}(t) \in \mathbb{R}^{4\times 1}$ denotes the resultant force of external disturbances and internal uncertain dynamic force, $\Upsilon(t)$ is the energy loss from viscous friction shown as

$$\Upsilon(t) = \frac{1}{2}\Upsilon_\theta\sum_{i=1}^{4}\varphi_i^2(t)$$

where $\Upsilon_\theta \in \mathbb{R}$ is the viscous friction coefficient of the Mecanum wheel. The dynamic model (6) is further obtained as

$$M\dot{\Omega}_v(t) + \Upsilon_\theta\Omega_v(t) = \Gamma_v(t) + \Gamma_{dv}(t)$$

with

$$M = \begin{bmatrix} c & -b & b & a-b \\ -b & c & a-b & b \\ b & a-b & c & -b \\ a-b & b & -b & c \end{bmatrix}$$

where $a = mR^2/8$, $b = I_zR^2/(16(l_x + l_y)^2)$, $c = a + b + I_w$. It is obtained that M is a nonsingular matrix due to full rank. Furthermore, a conversion relationship between the body velocity $V_v(t)$ and the wheel angular velocity $\Omega_v(t)$ is given as $\Omega_v(t) = JV_v(t)$, where

$$J = \frac{1}{R}\begin{bmatrix} 1 & 1 & -l_x - l_y \\ 1 & -1 & -l_x - l_y \\ 1 & 1 & l_x + l_y \\ 1 & -1 & l_x + l_y \end{bmatrix}.$$

A generalized inverse matrix of J is expressed as

$$J^+ = \frac{R}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ \frac{-1}{l_x+l_y} & \frac{-1}{l_x+l_y} & \frac{1}{l_x+l_y} & \frac{1}{l_x+l_y} \end{bmatrix}$$

Based on the dynamic model (6), a dynamic model of the virtual OMR is obtained as

$$MJ\dot{V}_v(t) + \Upsilon_\theta JV_v(t) = \Gamma_v(t) + \Gamma_{dv}(t) \tag{7}$$

By the same way, a dynamic model of the physical OMR is expressed as

$$MJ\dot{V}_p(t) + \Upsilon_\theta JV_p(t) = \Gamma_p(t) + \Gamma_{dp}(t) \tag{8}$$

where $\Gamma_p(t)$ is the control torque vector of four physical Mecanum wheels, $\Gamma_{dp}(t)$ denotes the resultant force of disturbances from the physical OMR.

C. Preliminaries

Assumption 1: For the physical OMR, the total disturbance $\Gamma_{dp}(t)$ which contains internal uncertain dynamics and external disturbances is bounded and continuously differentiable over time, and the derivative of $\Gamma_{dp}(t)$ is also bounded. As the digital mirror, the virtual OMR ensures that $\Gamma_{dv}(t)$ is bounded and $\dot{\Gamma}_{dp}(t)$ exists and is differentiable.

Remark 2: The physical OMR is taken as an example to analyze correctness of Assumption 1. Note that force interaction is inevitable for the physical OMR and physical environment. In this paper, uncertain dynamics comes from irregular structure of Mecanum wheels, external disturbances are due to uneven road in physical environment and skidding among OMR's four drive wheels, which result in imprecise trajectory tracking. A force from the road and acting on a tire may be changed by the external disturbances and uncertain dynamics which are the total disturbance for the physical OMR. The force is limited and usually changes smoothly over time for the reason of that main material of tires is elastic rubber whose elastic deformation is a

Authorized licensed use limited to: Universidad Federal de Pernambuco. Downloaded on July 31,2026 at 16:24:55 UTC from IEEE Xplore. Restrictions apply.

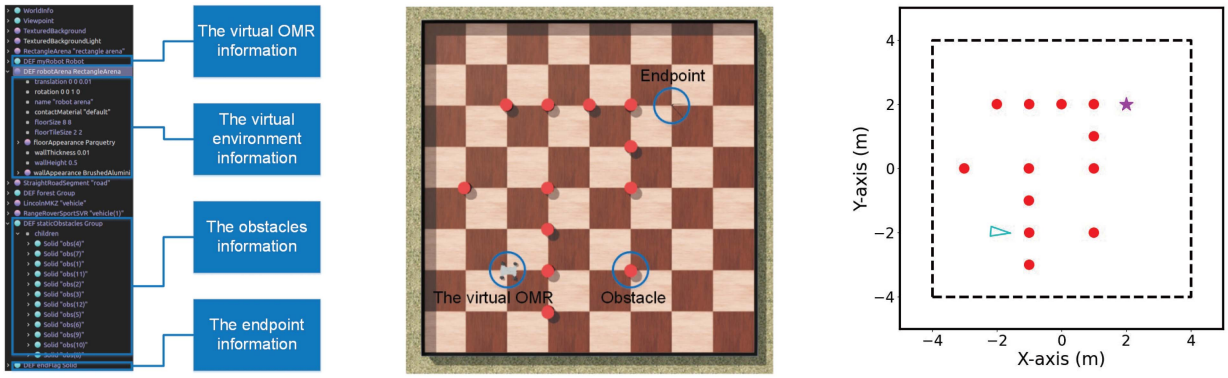


Fig. 4. The virtual space map information and grip map. (a) Map information in Webots. (b) The virtual environment. (c) The grip map and global path.

slow and continuous process. That is, the total disturbance acting on the physical OMR is bounded and continuously differentiable over time, and the derivative of the total disturbance is also bounded in application. In the following, an integral sliding mode controller is designed to achieve synchronized control under the influence of total disturbances.

IV. DT-Based Autonomous Navigation and Control

A. Global Path Planning

After an endpoint is set in the virtual environment of Webots, global path planning program firstly build a grid map of the virtual environment automatically for preparation of the Astar algorithm. Based on the grid map, the Astar algorithm is used to acquire the global path from the virtual OMR to the endpoint.

1) Grid Map Building: As shown in Fig. 4(a), map information of the virtual environment containing environmental boundaries, obstacles coordinates and sizes, the virtual OMR information and endpoint coordinate is intuitive and easily accessible in Webots. By interacting with Webots, the map information is directly read in program to build a 2D grid map. The virtual environment and the built grid map are illustrated in Fig. 4(b) and (c), respectively. Note that grid lines in Fig. 4(c) are hidden for readability and clarity since each square width of the built grid map is 0.1m. As shown in 4(c), black dashed line represents boundary of the map, red solid circles denote position and size of the obstacles, purple pentagram is the endpoint, blue triangle indicates position and orientation of the virtual OMR.

2) Astar Algorithm: The Astar algorithm is a heuristic-based path planning algorithm [17]. To obtaining search direction of the Astar algorithm in the grip map, a path length evaluation function is given as

$$F(n) = G(n) + H(n)$$

where $F(n)$ represents the estimated cost from the nth node to the endpoint, $G(n)$ indicates the actual cost of the planned path from the starting point to the current node n, the heuristic function $H(n)$ is used to estimate cost of the current node n to the endpoint. In this paper, $H(n)$ is obtained by

$$H(n) = \sqrt{(x_n - x_e)^2 + (y_n - y_e)^2}$$

where (x_n, y_n) and (x_e, y_e) represent the coordinates of the current node n and the endpoint in the grip map, respectively. By the Astar algorithm, a global path from the virtual OMR to the endpoint is planned in the virtual space.

B. Local Path Planning

On the basis of the global path planned by an A-star algorithm, the DWA is employed to calculate the optimal collision free velocity at which the virtual OMR achieves the endpoint [21]. Concrete implementation of the DWA mainly includes three parts: Velocity space determination, motion trajectories prediction and trajectories evaluation.

1) Velocity Space Determination: The DWA describes local path planning problem as an optimization problem with constraints in velocity space. Owing to the DT system, a velocity constraint of the physical OMR shows a velocity range of the virtual OMR. Therefore, it is a top priority to determine the velocity range of the physical OMR from the following three aspects.

Firstly, the velocity range of the physical OMR is constrained by operating environment and rated motor speed as

$$\underline{v} \le v_p(t) \le \overline{v},\ \underline{\nu} \le \nu_p(t) \le \overline{\nu},\ \underline{\omega} \le \omega_p(t) \le \overline{\omega} \tag{9}$$

where $\underline{v}$, $\underline{\nu}$ and $\underline{\omega}$ represent the minimum of $v_p(t)$, $\nu_p(t)$ and $\omega_p(t)$, respectively, $\overline{v}$, $\overline{\nu}$ and $\overline{\omega}$ denote maximum of $v_p(t)$, $\nu_p(t)$ and $\omega_p(t)$, respectively. According to the velocity range (9), part one velocity space of the virtual OMR is expressed as

$$\mathbb{V}_1 = \{(v, \nu, \omega) | v \in [\underline{v}, \overline{v}] \cap \nu \in [\underline{\nu}, \overline{\nu}] \cap \omega \in [\underline{\omega}, \overline{\omega}]\} \tag{10}$$

where (v, ν, ω) is an element of the set $\mathbb{V}_1$.

Secondly, the maximum acceleration of the physical OMR in all directions is constrained by performance of the four wheel motors. The velocity range of the physical OMR at next time $t + \Delta t$ is determined by the current velocity $V_p(t)$ and the maximum acceleration as

$$\begin{cases} v_p(t) + \check{a}_v \Delta t \le v_p(t) \le v_p(t) + \hat{a}_v \Delta t,\ \check{a}_v \le 0 \\ \nu_p(t) + \check{a}_\nu \Delta t \le \nu_p(t) \le \nu_p(t) + \hat{a}_\nu \Delta t,\ \check{a}_\nu \le 0 \\ \omega_p(t) + \check{a}_\omega \Delta t \le \omega_p(t) \le \omega_p(t) + \hat{a}_\omega \Delta t,\ \check{a}_\omega \le 0 \end{cases} \tag{11}$$

Authorized licensed use limited to: Universidad Federal de Pernambuco. Downloaded on July 31,2026 at 16:24:55 UTC from IEEE Xplore. Restrictions apply.

where $\hat{a}_v$ and $\check{a}_v$ represent the maximum acceleration and deceleration of $v_p(t)$, respectively, $\hat{a}_\nu$ and $\check{a}_\nu$ indicate the maximum acceleration and deceleration of $\nu_p(t)$, respectively, $\hat{a}_\omega$ and $\check{a}_\omega$ denote the maximum acceleration and deceleration of $\omega_p(t)$, respectively. According to the velocity range (11), part two velocity space of the virtual OMR is obtained as

$$\begin{aligned}\mathbb{V}_2 = \{(v,\nu,\omega)\,|\,v \in [v_v(t)+\check{a}_v\Delta t, v_v(t)+\hat{a}_v\Delta t] \\ \cap\, \nu \in [\nu_v(t)+\check{a}_\nu\Delta t, \nu_v(t)+\hat{a}_\nu\Delta t] \\ \cap\, \omega \in [\omega_v(t)+\check{a}_\omega\Delta t, \omega_v(t)+\hat{a}_\omega\Delta t]\}\end{aligned} \tag{12}$$

Lastly, the current velocity of the physical OMR should be able to decelerate to 0 before hitting an obstacle under the maximum deceleration for safety. Due to omnidirectional movement characteristics of OMRs, the distance between the physical OMR and an obstacle at current time t should be expressed with the Euclidean distance as $D(t)$. The safety velocity of the physical OMR is further limited as

$$\begin{cases} v_p(t) \le \sqrt{2D(t)|\check{a}_v|} \\ \nu_p(t) \le \sqrt{2D(t)|\check{a}_\nu|} \\ \omega_p(t) \le \sqrt{2D(t)|\check{a}_\omega|} \end{cases} \tag{13}$$

According to the velocity range (13), part one velocity space of the virtual OMR is shown as

$$\begin{aligned}\mathbb{V}_3 = \{(v,\nu,\omega)\,|\,v \le \sqrt{2D(t)|\check{a}_v|} \\ \cap\, \nu \le \sqrt{2D(t)|\check{a}_\nu|} \cap \omega \le \sqrt{2D(t)|\check{a}_\omega|}\}\end{aligned} \tag{14}$$

The velocity space of the virtual OMR is intersection of the partial velocity space (10), (12) and (14) as follows

$$\mathbb{V} = \mathbb{V}_1 \cap \mathbb{V}_2 \cap \mathbb{V}_3 \tag{15}$$

2) Motion Trajectories Prediction: Motion trajectories prediction is that multiple sets of velocities sampled in the definitive velocity space (15) are input to the kinematic model (1) to predict multiple sets of future trajectories for a period of time. However, the sampling rule is difficult to choose. Let $(\Delta v, \Delta\nu, \Delta\omega)$ represents sampling velocity increment of the velocity space (15). Excessive sampling velocity increment results in too few predicted trajectories to obtain the optimal trajectory. Too small sampling velocity increment inevitably lead to complex calculation and affect real-time performance. Based on the acceleration constraint of the partial velocity space (12), the sampling velocity increment of the velocity space (15) is given as

$$\begin{cases} \Delta v = (\hat{a}_v - \check{a}_v)\Delta t/k \\ \Delta\nu = (\hat{a}_\nu - \check{a}_\nu)\Delta t/k \\ \Delta\omega = (\hat{a}_\omega - \check{a}_\omega)\Delta t/k \end{cases} \tag{16}$$

where $k \in \mathbb{N}^+$ is used to adjust size of the sampling velocity increment. By the sampling velocity increment (16), multiple sets of (v_i, ν_i, ω_i) with $i \in \mathbb{N}^+$ are sampled in the velocity space (15).

Set two prediction trajectories on the velocity (v_i, ν_i, ω_i) and the prediction time as PT_i and T, respectively. The points of PT_i at the sequence $[t, t+\Delta t, t+2\Delta t, \ldots, t+T]$ are inferred by the initial state $Q_v(t)$ and the sampling velocity (v_i, ν_i, ω_i). Inference relationship is obtained by linearizing the kinematic model (1) of the virtual OMR as

$$Q_v(t+\Delta t) = Q_v(t) + \Delta t * T(\theta_v(t)) \begin{bmatrix} v_i \\ \nu_i \\ \omega_i \end{bmatrix} \tag{17}$$

Note that a prediction trajectory should be abandoned once the trajectory point coincides with the obstacle or the boundary.

3) Trajectories Evaluation: An evaluation function is designed to select an optimal trajectory from the multiple sets of the predicted trajectories as follows

$$G(PT_i) = \alpha Goal(PT_i) + \beta Obs(PT_i) + \gamma Vel(PT_i) \tag{18}$$

where α, β and γ represent weighting coefficients, $Goal(PT_i)$, $Obs(PT_i)$ and $Vel(PT_i)$ denote the goal evaluation function, the obstacle evaluation function and the velocity evaluation function of PT_i, respectively. The evaluation function (18) aims to select faster collision free velocity towards the endpoint at current time t.

$Goal(PT_i)$ is used to evaluate degree of deviation between PT_i and the global path. $Goal(PT_i)$ is Euclidean distance between the last point of PT_i and a dynamic preview point of the global path. The smaller the distance is, the larger $Goal(PT_i)$ is. Update rule of the dynamic preview point is that the Euclidean distance between the dynamic preview point and the current position of the virtual OMR is always greater than a threshold. When reaching the endpoint, the dynamic preview point remains equal to the endpoint.

$Obs(PT_i)$ is used to evaluate obstacle avoidance ability of PT_i. Let $dist(PT_i)$ represent Euclidean distance between PT_i and the obstacle which is closest to PT_i. The larger $dist(PT_i)$ is, the larger $Obs(PT_i)$ is. However, too large $dist(PT_i)$ has adverse effects on $G(PT_i)$ when there are no obstacle around PT_i. To balance the shortest trajectory and obstacle avoidance, $Obs(PT_i)$ is obtained as

$$Obs(PT_i) = \begin{cases} dist(PT_i),\ dist(PT_i) \le L \\ L,\ dist(PT_i) > L \end{cases}$$

where $L \in \mathbb{N}^+$ is the adjustable threshold of obstacle avoidance.

$Vel(PT_i)$ is used to evaluate (v_i, ν_i, ω_i) of PT_i. The larger velocity is, the larger $Vel(PT_i)$ is. Therefore, $Vel(PT_i)$ is obtained by

$$Vel(PT_i) = \sqrt{v_i^2 + \nu_i^2}$$

Among all predicted trajectories at current time t, the velocity with the largest $G(PT_i)$ is given to the virtual OMR as the control input $U_v(t)$ at time t. By continuously iterating to obtain $U_v(t)$, the virtual OMR will reach the endpoint without collision in a shorter time.

C. Physical-Virtual Motion Synchronization

The physical-virtual motion synchronization is a critical step to realize the virtual trajectory in the physical space after obtaining the optimal virtual navigation trajectory as shown in Fig. 1.

Authorized licensed use limited to: Universidad Federal de Pernambuco. Downloaded on July 31,2026 at 16:24:55 UTC from IEEE Xplore. Restrictions apply.

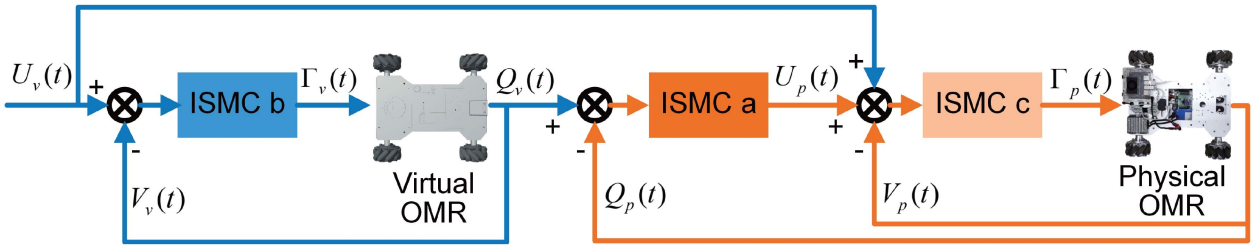


Fig. 5. Physical-virtual motion synchronization framework.

A framework of the physical-virtual motion synchronization method is designed based on feedforward control and feedback control as illustrated in Fig. 5. In Fig. 5, the control input $U_v(t)$ is input to both the virtual OMR and the physical OMR. In [11], implementation of the physical-virtual motion synchronization will only rely on $U_v(t)$ if the virtual space is perfect modeling of the physical space in all aspects. However, it is inevitable for errors in the virtual space modeling since internal and external disturbances of the physical OMR and uneven ground disturbances are hard to obtain accurately. Therefore, an integral sliding mode controller marked as "ISMC a" in Fig. 5 is proposed to correct position and angle errors between the virtual OMR and the physical OMR while $U_v(t)$ serves as a feedforward to input the physical OMR. For keeping up with expected velocities, two integral sliding mode controllers marked as "ISMC b" and "ISMC c" in Fig. 5 are designed to control torques of the virtual OMR and the physical OMR, respectively.

1) Integral Sliding Mode Controller A): Based on [9], $E(t) = Q_v(t) - Q_p(t) = [e_x(t), e_y(t), e_\theta(t)]^{\mathrm{T}}$ express the state error between the virtual OMR and the physical OMR. A state error system is obtained as

$$\dot{E}(t) = \dot{Q}_v(t) - \dot{Q}_p(t) \tag{19}$$

Design goal of the integral sliding mode controller a) is to stabilize the state error system (19). An integral sliding mode surface is proposed as

$$S(t) = E(t) + K_1 \int_0^t E(\xi)d\xi - E(0) \tag{20}$$

where $S(t) \in \mathbb{R}^{3\times 1}$ represents the sliding mode surface vector corresponding to the state error vector $E(t)$, $K_1 \in \mathbb{R}^{3\times 3}$ is the adjustable parameter matrix with nonnegative elements, $E(0)$ represents the initial state error. The integral sliding mode surface $S(t)$ is used such that the state $E(t)$ is on the sliding mode surface at initial moment. Derivative of the integral sliding mode surface (20) is calculated as

$$\begin{aligned}\dot{S}(t) &= \dot{E}(t) + K_1 E(t)\\ &= \dot{Q}_v(t) - T(\theta_p(t))V_p(t) + K_1 E(t)\end{aligned} \tag{21}$$

Then a convergence rate is designed as

$$\dot{S}(t) = -K_2 S(t) - K_3 \mathrm{F}(S(t)) \tag{22}$$

where $K_2, K_3 \in \mathbb{R}^{3\times 3}$ are the adjustable positive parameter matrices with nonnegative elements, $\mathrm{F}(S(t))$ is shown as

$$\mathrm{F}(S(t)) = [\mathrm{atan}(S(t)[1]), \mathrm{atan}(S(t)[2]), \mathrm{atan}(S(t)[3])]^{\mathrm{T}}$$

The arctangent function $\mathrm{atan}(\cdot)$ is beneficial for eliminating chattering in the integral sliding mode control. According to the equations (19), (21) and (22), it is obtained that

$$\dot{Q}_p(t) = \dot{Q}_v(t) + K_1 E(t) + K_2 S(t) + K_3 \mathrm{F}(S(t)) \tag{23}$$

Considering the kinematic model (1) and the (23), the integral sliding mode controller a) marked as "ISMC a" in Fig. 5 is designed as

$$\begin{aligned}U_p(t) = T^{-1}(\theta_p(t))(&\dot{Q}_v(t) + K_1 E(t)\\ &+ K_2 S(t) + K_3 f(S(t)))\end{aligned} \tag{24}$$

The physical-virtual motion synchronization is guaranteed by the integral sliding mode controller (24).

Theorem 1: Consider the state error system (19) and the integral sliding mode controller (24). If the body velocity $V_p(t)$ is equal to the control signal $U_p(t)$ at time $t > 0$, the state error system (19) stabilizes by choosing the parameter matrices K_1, K_2 and K_3 with nonnegative elements. That is, the physical OMR maintains motion synchronization with the virtual OMR.

Proof: A Lyapunov function is given as

$$\mathbf{V}_1(t) = \frac{1}{2} S^{\mathrm{T}}(t) S(t)$$

According to the convergence rate (22), the derivative of $\mathbf{V}_1(t)$ is calculated as

$$\begin{aligned}\dot{\mathbf{V}}_1(t) &= \frac{1}{2}\left(\dot{S}_1^{\mathrm{T}}(t)S(t) + S^{\mathrm{T}}(t)\dot{S}_1(t)\right)\\ &= S^{\mathrm{T}}(t)\dot{S}_1(t)\\ &= -S^{\mathrm{T}}(t)K_2 S(t) - S^{\mathrm{T}}(t)K_3 \mathrm{F}(S(t))\\ &\le 0\end{aligned}$$

That is, $S(t)$ eventually stabilizes to zero although the time-varying virtual trajectory may cause that state deviates from the sliding mode surface (20). Therefore, the state error system (19) is asymptotically stable.

2) Integral Sliding Mode Controller B) and C): The virtual OMR is taken as an example to design the integral sliding mode controller b). Let $\Phi_v(t) = U_v(t) - V_v(t)$ represent the error vector between expected input velocity and current state of the virtual OMR. According to the dynamic model (7), a velocity tracking error system is obtained as

$$\begin{aligned}\dot{\Phi}_v(t) &= \dot{U}_v(t) - \dot{V}_v(t)\\ &= \dot{U}_v(t) - J^{+}M^{-1}\left(\Gamma_v(t) + \Gamma_{dv}(t) - \Upsilon_\theta J V_v(t)\right)\end{aligned} \tag{25}$$

Authorized licensed use limited to: Universidad Federal de Pernambuco. Downloaded on July 31,2026 at 16:24:55 UTC from IEEE Xplore. Restrictions apply.

To reduce impact of disturbances on the velocity tracking error system (25), integral sliding control is considered to design the wheel torque controller. An integral sliding mode surface is given as

$$S_v(t) = \Phi_v(t) + \lambda_{1v} \int_0^t \Phi_v(\xi) d\xi - \Phi_v(0) \tag{26}$$

where $S_v(t) \in \mathbb{R}^{3\times 1}$ is the sliding mode surface vector, $\lambda_{1v} \in \mathbb{R}^{3\times 3}$ is the adjustable parameter matrix with nonnegative elements of $S_v(t)$, $\Phi_v(0)$ represents the initial state error. The integral sliding mode surface $S_v(t)$ is given so that $\Phi_v(t)$ is on the sliding mode surface at initial moment. One has that

$$\begin{aligned} \dot{S}_v(t) &= \dot{\Phi}_v(t) + \lambda_{1v}\Phi_v(t) \\ &= -J^+ M^{-1} \left(\Gamma_v(t) + \Gamma_{dv}(t) - \Upsilon_\theta J V_v(t)\right) \\ &\quad + \dot{U}_v(t) + \lambda_{1v}\Phi_v(t) \end{aligned} \tag{27}$$

A convergence rate is designed as

$$\dot{S}_v(t) = -\lambda_{2v} S_v(t) - \lambda_{3v} \mathrm{F}_v(S_v(t)) \tag{28}$$

where $\lambda_{2v}, \lambda_{3v} \in \mathbb{R}^{3\times 3}$ are the adjustable parameter matrices with nonnegative elements, $\mathrm{F}_v(S_v(t))$ is shown as

$$\mathrm{F}_v(S_v(t)) = [\mathrm{atan}(S_v(t)[1]), \mathrm{atan}(S_v(t)[2]), \mathrm{atan}(S_v(t)[3])]^{\mathrm{T}}$$

According to the (27) and (28), the integral sliding mode controller b) marked as "ISMC b" in Fig. 5 is given as

$$\begin{aligned} \Gamma_v(t) = L_v \mathrm{Sig}(S_v(t)) + MJ(\dot{U}_v(t) + \lambda_{1v}\Phi_v(t) \\ + \lambda_{2v} S_v(t) + \lambda_{3v} \mathrm{F}_v(S_v(t))) \end{aligned} \tag{29}$$

where $\mathrm{Sig}(S_v(t))$ is the function vector and $L_v \in \mathbb{R}^{3\times 1}$ is the constant vector, i.e.,

$$\begin{aligned} \mathrm{Sig}(S_v(t)) = [\mathrm{sign}(S_v(t)[1]), \mathrm{sign}(S_v(t)[2]), \mathrm{sign}(S_v(t)[3])]^{\mathrm{T}} \\ L_v[j] > \left|\left(\Gamma_{dv}(t) - \Upsilon_\theta J V_v(t)\right)[j]\right|, \; j \in 1, 2, 3 \end{aligned}$$

Convergence of the velocity tracking error system (25) is guaranteed by the integral sliding mode controller (29).

Theorem 2: Consider the integral sliding mode controller (29). The velocity tracking error $\Phi_v(t)$ of the virtual OMR asymptotically converges to zero by choosing the positive parameter vector L_v and the parameter matrices λ_{1v}, λ_{2v} and λ_{3v} with nonnegative elements.

Proof: A Lyapunov function is selected as

$$\mathbf{V}_2(t) = \frac{1}{2} S_v^{\mathrm{T}}(t) S_v(t)$$

Considering the (27), it is obtained that

$$\begin{aligned} \dot{\mathbf{V}}_2(t) &= \frac{1}{2}\left(\dot{S}_v^{\mathrm{T}}(t) S_v(t) + S_v^{\mathrm{T}}(t)\dot{S}_v(t)\right) \\ &= S_v^{\mathrm{T}}(t)\dot{S}_v(t) \\ &= S_v^{\mathrm{T}}(t)(\dot{U}_v(t) + \lambda_{1v}\Phi_v(t) \\ &\quad - J^+ M^{-1}(\Gamma_v(t) + \Gamma_{dv}(t) - \Upsilon_\theta J V_v(t))) \end{aligned}$$

According to the integral sliding mode controller (29), $\dot{\mathbf{V}}_2(t)$ is further calculated as

$$\begin{aligned} \dot{\mathbf{V}}_2(t) &= S_v^{\mathrm{T}}(t)(-\lambda_{2v} S_v(t) - \lambda_{3v}\mathrm{F}_v(S_v(t)) \\ &\quad - J^+ M^{-1}(L_v \mathrm{Sig}(S_v(t)) + \Gamma_{dv}(t) - \Upsilon_\theta J V_v(t))) \\ &= -S_v^{\mathrm{T}}(t)\lambda_{2v} S_v(t) - S_v^{\mathrm{T}}(t)\lambda_{3v}\mathrm{F}_v(S_v(t)) - S_v^{\mathrm{T}}(t) \\ &\quad \times J^+ M^{-1}(L_v \mathrm{Sig}(S_v(t)) + \Gamma_{dv}(t) - \Upsilon_\theta J V_v(t)) \end{aligned}$$

It is obtained that L_v exists due to Assumption 1. Therefore, $\dot{\mathbf{V}}_2(t) \leq 0$. That is, $S_v(t)$ eventually stabilizes to zero although the time-varying virtual trajectory causes state to deviate from the sliding mode surface (27). Therefore, the state error system (25) is asymptotically stable.

By the same way, the integral sliding mode controller c) marked as "ISMC c" in Fig. 5 is obtained as

$$\begin{aligned} \Gamma_p(t) = L_p \mathrm{Sig}(S_p(t)) + MJ(\dot{U}_v(t) + \dot{U}_p(t) \\ + \lambda_{1p}\Phi_p(t) + \lambda_{2p} S_p(t) + \lambda_{3p}\mathrm{F}_p(S_p(t))) \end{aligned} \tag{30}$$

where $\Phi_p(t) = U_v(t) + U_p(t) - V_p(t)$ is the error vector between the expected input velocity and the current state of the physical OMR as shown in Fig. 5, $S_p(t) \in \mathbb{R}^{3\times 1}$ is the integral sliding mode surface as

$$S_p(t) = \Phi_p(t) + \lambda_{1p} \int_0^t \Phi_p(\xi) d\xi - \Phi_p(0),$$

$\mathrm{F}_p(S_p(t))$and $\mathrm{Sig}(S_p(t))$ are the function vectors, $L_p \in \mathbb{R}^{3\times 1}$ is a constant vector as

$$\begin{aligned} \mathrm{F}_p(S_p(t)) &= [\mathrm{atan}(S_p(t)[1]), \mathrm{atan}(S_p(t)[2]), \mathrm{atan}(S_p(t)[3])]^{\mathrm{T}} \\ \mathrm{Sig}(S_p(t)) &= [\mathrm{sign}(S_p(t)[1]), \mathrm{sign}(S_p(t)[2]), \mathrm{sign}(S_p(t)[3])]^{\mathrm{T}} \\ L_p[j] &> \left|\left(\Gamma_{dp}(t) - \Upsilon_\theta J V_p(t)\right)[j]\right|, \; j \in 1, 2, 3 \end{aligned}$$

where $\lambda_{1p}, \lambda_{2p}, \lambda_{3p} \in \mathbb{R}^{3\times 3}$ is the adjustable parameter matrices with nonnegative elements. With the integral sliding mode controller (30), it is guaranteed that the velocity tracking error $\Phi_p(t)$ asymptotically converges to zero.

Corollary 1: Consider the integral sliding mode controller (30). The velocity tracking error $\Phi_p(t)$ of the physical OMR asymptotically converges to zero by choosing the positive parameter vector L_p and the parameter matrices λ_{1p}, λ_{2p} and λ_{3p} with nonnegative elements.

Remark 3: In this paper, DT is established in virtual environment where there exists high computational power based on the server. By the high computational power, the Astar algorithm and the DWA are integrated to solve a trajectory iteratively with high precision. Note that traditional DWAs usually get stuck in local optima in obstacle avoidance, which has been avoided by the proposed method in this paper. The $Goal(PT_i)$ term in the evaluation function (18) is used to quantify the deviation between PT_i and $globalpath$ for replacing the original $Heading(PT_i)$ orientation evaluation function in traditional DWAs. Based on the DT system, the autonomous navigation method is deployed on an upper-level server to enhance trajectories solving accuracy by leveraging the computational power. Then navigation tasks of the physical OMR are accomplished by the physical-virtual motion synchronization method based on the integral sliding mode controllers (29) and (30). For other mobile robots with non-holonomics constraints, the Astar algorithm and the DWA should be modified to made available for specific robots before application.

Authorized licensed use limited to: Universidad Federal de Pernambuco. Downloaded on July 31,2026 at 16:24:55 UTC from IEEE Xplore. Restrictions apply.



Fig. 6. The physical environment.

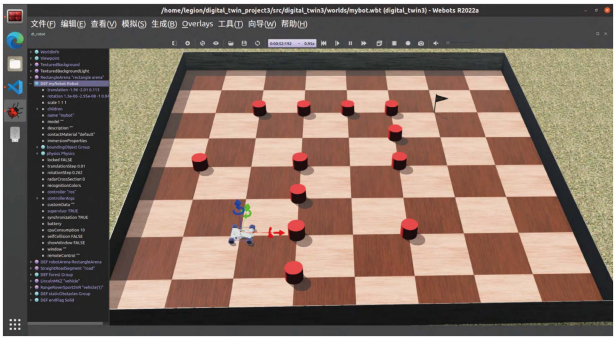


Fig. 7. Webots interface of the virtual space.

TABLE I
PARAMETERS OF THE LOCAL PATH PLANNING

Parameters	Values
$\overline{v}$, $\underline{v}$	1 m/s, −1 m/s
$\overline{\nu}$, $\underline{\nu}$	1 m/s, −1 m/s
$\overline{\omega}$, $\underline{\omega}$	1.57 rad/s, −1.57 rad/s
$\hat{a}_v$, $\check{a}_v$	0.5 m/s^2, −0.5 m/s^2
$\hat{a}_\nu$, $\check{a}_\nu$	0.5 m/s^2, −0.5 m/s^2
$\hat{a}_\omega$, $\check{a}_\omega$	1.57 rad/s^2, 1.57 rad/s^2
Δt, T	0.1s, 2.5s
α, β, γ	1.0, 1.0, 0.3
k	4
L	0.6 m

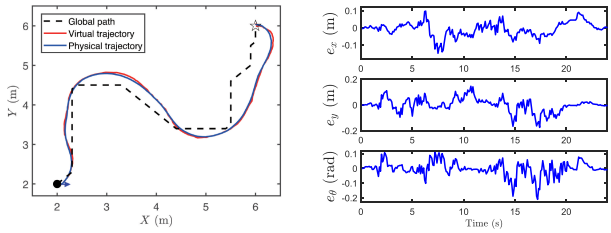
V. EXPERIMENTAL EVALUATION

In this paper, an experiment is carried out to show effectiveness of the proposed autonomous navigation and control method based on DT. A physical environment is arranged to contain the physical OMR and obstacles composed of warning cones in Fig. 6. In Fig. 6, base of the warning cone is considered as a circle with a radius of 0.3 m, four UWB anchors are placed on four corners of a 8 m× 8m square to determine the physical world coordinate system. Webots interface of the virtual space corresponding to the physical space is illustrated in Fig. 7. In Fig. 7, red cylinders in the virtual space have the same position and base radius as the warning cones in the physical space. In the physical-virtual synchronization experiment, the starting coordinates of the physical OMR and the virtual OMR are all set to (2,2), while navigation endpoint coordinates are all set to (6,6).

All parameters of the local path planning are listed in Table I. Both the model and controller parameters are incorporated in Table II. Global path, the trajectories of the virtual OMR and the physical OMR are shown in Fig. 8(a). In Fig. 8(a), the

TABLE II
PARAMETERS OF THE MODEL AND CONTROLLERS

Parameters	Values
K_1	diag$\{0.9, 1.1, 0.3\}$
K_2	diag$\{1.8, 2.3, 1.2\}$
K_3	diag$\{0.7, 0.7, 0.5\}$
L_v, L_p	$[4, 4, 1.5]^{\mathrm{T}}$, $[3, 3, 1.8]^{\mathrm{T}}$
λ_{1v}, λ_{1p}	diag$\{1.2, 1.1, 1.0\}$, diag$\{0.9, 1.1, 0.8\}$
λ_{2v}, λ_{2p}	diag$\{0.8, 0.7, 1.2\}$, diag$\{1.6, 1.8, 1.0\}$
λ_{3v}, λ_{3p}	diag$\{2.7, 2.5, 1.5\}$, diag$\{2.5, 2.3, 0.9\}$
m, R	2.53kg, 0.05m
l_x, l_y	0.156m, 0.176m
I_z, I_w	0.07kg·m^2, 0.01kg·m^2



(a) (b)

Fig. 8. Results of physical-virtual synchronization experiment. (a) Global path and trajectories. (b) The physical-virtual error.

start state is set to $[x_v(0), y_v(0), \theta_v(0)]^{\mathrm{T}} = [2, 2, 0]^{\mathrm{T}}$, the Astar algorithm plans a global path to the endpoint, the virtual OMR arrives at the virtual endpoint by DWA, the physical OMR arrives at the physical endpoint by the physical-virtual motion synchronization method. Feasibility of the proposed autonomous navigation and control method based on DT is shown in Fig. 8(a). The state error $E(t)$ between the virtual OMR and the physical OMR is illustrated in Fig. 8(b). The RMSE is used to describe the tracking error between the virtual path and the physical path, which is only 0.061 m in this method. Control accuracy and robustness of the designed integral sliding mode controller (24) are indicated in Fig. 8(b). Screenshots of experiment process is shown in Fig. 9. Let $\Phi_v(t) = [\varepsilon_{vv}(t), \varepsilon_{v\nu}(t), \varepsilon_{v\omega}(t)]^{\mathrm{T}}$ and $\Phi_p(t) = [\varepsilon_{pv}(t), \varepsilon_{p\nu}(t), \varepsilon_{p\omega}(t)]^{\mathrm{T}}$. The velocity tracking errors $\Phi_v(t)$ of the virtual OMR and $\Phi_p(t)$ of the physical OMR are shown in Fig. 10(a) and (b), respectively. In Fig. 10(a) and (b), effectiveness and celerity are shown for the integral sliding mode controllers (29) and (30), respectively.

After the physical-virtual synchronization experiment, different random tasks are set to show effectiveness of the navigation method proposed in this paper. Each task includes a pair of starting coordinate and ending coordinate, and a task group includes several tasks. Two task groups are generated randomly in the map, and each task group contains 10 tasks. Based on the navigation algorithm and control algorithms, all 20 tasks are finished with no collisions. Trajectories for all tasks are shown in Fig. 11(a) and (b). In Fig. 11, effectiveness and applicability are shown for the autonomous navigation method.

Remark 4: Table III shows a comparison of different navigation algorithms.In [35] and [36], many imprecise computation issues have been avoided by the proposed method in this paper. Moreover, the proposed method in this paper demonstrates strong generality without relying on extensive datasets

Authorized licensed use limited to: Universidad Federal de Pernambuco. Downloaded on July 31,2026 at 16:24:55 UTC from IEEE Xplore. Restrictions apply.

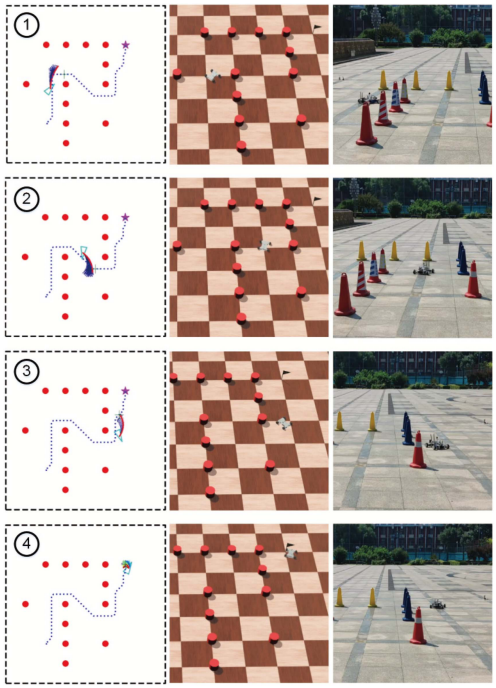


Fig. 9. Screenshots of experiment process.

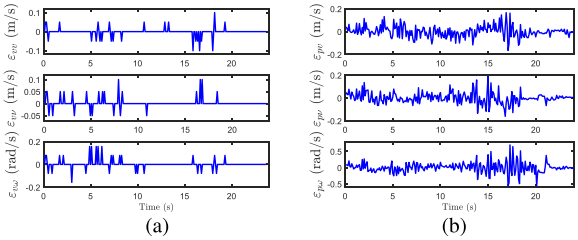


Fig. 10. The velocity tracking error results. (a) The virtual OMR. (b) The physical OMR.

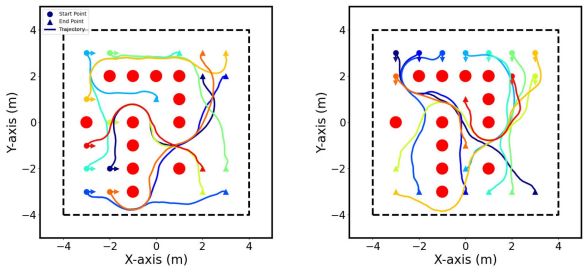


Fig. 11. Trajectory results for random tasks experiment. (a) The first task group. (b) The second task group.

TABLE III
COMPARATIVE ANALYSIS OF THE ALGORITHMS

Related work	No simplified calculation	Not dependent on datasets	No training process
[35], [36]		✓	✓
[37], [38]	✓		
This paper	✓	✓	✓

and some prolonged algorithm training processes in contrast to deep reinforcement learning [37] and [38]. Based on the DT system, the proposed autonomous navigation method can get a more feasible and safe trajectory for OMRs. The theoretical derivations and experimental validations on the proposed method in this paper have addressed the detailed comparison with existing related algorithms. Compared with the existing related algorithms in [35], [36], [37], [38], the proposed method in this paper is characterized by precise computation and high applicability. So we do not show performance metrics which are quantitative in this paper.

VI. CONCLUSION

In this paper, a DT-based autonomous navigation and control method has been proposed for an OMR. Based on the constructed four-dimensional DT system, an Astar algorithm and a dynamic windows approach have been fused to handle global and local path planning in virtual environment. An integral sliding mode controller is designed for the physical-virtual motion synchronization. Then two integral sliding mode controllers have been proposed for the physical OMR and the virtual OMR to make the velocity tracking errors be convergent. Finally, experiment results have shown effectiveness of the DT-based autonomous navigation and control method. In future work, the autonomous navigation method will be improved to deal with dynamic obstacle avoidance utilizing the DT system.

REFERENCES

[1] I. Chio, K. Ruan, Z. Wu, K. I. Wong, L. M. Tam, and Q. Xu, "Design and autonomous navigation of a new indoor disinfection robot based on disinfection modeling," *IEEE Trans. Automa. Sci. Eng.*, vol. 20, no. 1, pp. 649–661, Jan. 2023.

[2] Y. Mo, Z. Sun, and C. Yu, "Eventtube: An artificial intelligent edge computing based event aware system to collaborate with individual devices in logistics systems," *IEEE Trans. Ind. Inform.*, vol. 19, no. 2, pp. 1823–1832, Feb. 2023.

[3] S. Lin, A. Liu, J. Wang, and X. Kong, "An intelligence-based hybrid PSO-SA for mobile robot path planning in warehouse," *J. Comput. Sci.*, vol. 67, 2023, Art no. 101938.

[4] P. Pauwels, R. de Koning, B. Hendrikx, and E. Torta, "Live semantic data from building digital twins for robot navigation: Overview of data transfer methods," *Adv. Eng. Inform.*, vol. 56, 2023, Art no. 101959.

[5] Z. Zhou, X. Yang, H. Wang, and X. Zhang, "Digital twin with integrated robot-human/environment interaction dynamics for an industrial mobile manipulator," in *Proc. 2022 IEEE Int. Conf. Robot. Automat.*, 2022, pp. 5041–5047.

[6] T. Rantala, J. Ukko, M. Nasiri, and M. Saunila, "Shifting focus of value creation through industrial digital twins-from internal application to ecosystem-level utilization," *Technovation*, vol. 125, 2023, Art no. 102795.

[7] C. Zhang, L. Zhu, and C. Xu, "BSDP: Blockchain-based smart parking for digital-twin empowered vehicular sensing networks with privacy protection," *IEEE Trans. Ind. Inform.*, vol. 19, no. 5, pp. 7237–7246, May 2023.

[8] G. Shen, L. Lei, X. Zhang, Z. Li, S. Cai, and L. Zhang, "Multi-UAV cooperative search based on reinforcement learning with a digital twin driven training framework," *IEEE Trans. Veh. Technol.*, vol. 72, no. 7, pp. 8354–8368, Jul. 2023.

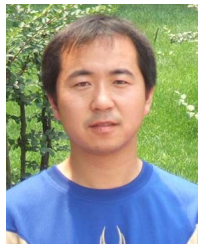
[9] H. Yang, F. Cheng, H. Li, and Z. Zuo, "Design and control of digital twin systems based on a unit level wheeled mobile robot," *IEEE Trans. Veh. Technol.*, vol. 73, no. 1, pp. 323–332, Jan. 2024.

[10] Z. Ren, J. Shi, and M. Imran, "Data evolution governance for ontology-based digital twin product lifecycle management," *IEEE Trans. Ind. Inform.*, vol. 19, no. 2, pp. 1791–1802, Feb. 2023.

[11] L. Zhao, Z. Nie, Y. Xia, and H. Li, "Virtual–physical tracking control for a car-like mobile robot based on digital twin technology," *IEEE Trans. Ind. Electron.*, vol. 71, no. 12, pp. 16348–16356, Dec. 2024.

Authorized licensed use limited to: Universidad Federal de Pernambuco. Downloaded on July 31,2026 at 16:24:55 UTC from IEEE Xplore. Restrictions apply.

[12] W. Jia, W. Wang, and Z. Zhang, "From simple digital twin to complex digital twin Part II: Multi-scenario applications of digital twin shop floor," *Adv. Eng. Inform.*, vol. 56, 2023, Art no. 101915.
[13] B. Li, Y. Liu, L. Tan, H. Pan, and Y. Zhang, "Digital twin assisted task offloading for aerial edge computing and networks," *IEEE Trans. Veh. Technol.*, vol. 71, no. 10, pp. 10863–10877, Oct. 2022.
[14] Y. Tang, Y. Wang, H. Tan, W. Peng, H. Xie, and X. Liu, "A digital twin-based intelligent robotic measurement system for freeform surface parts," *IEEE Trans. Instrum. Meas.*, vol. 72, 2023, Art. no. 7003013.
[15] C. Hu, L. Zhao, and G. Qu, "Event-triggered model predictive adaptive dynamic programming for road intersection path planning of unmanned ground vehicle," *IEEE Trans. Veh. Technol.*, vol. 70, no. 11, pp. 11228–11243, Nov. 2021.
[16] M. Lotfi et al., "A dijkstra-inspired graph algorithm for fully autonomous tasking in industrial applications," *IEEE Trans. Ind. Appl.*, vol. 57, no. 5, pp. 5448–5460, Sep./Oct. 2021.
[17] Z. Zhang, J. Jiang, J. Wu, and X. Zhu, "Efficient and optimal penetration path planning for stealth unmanned aerial vehicle using minimal radar cross-section tactics and modified A-star algorithm," *ISA Trans.*, vol. 134, pp. 42–57, 2023.
[18] S. Xie, J. Hu, P. Bhowmick, Z. Ding, and F. Arvin, "Distributed motion planning for safe autonomous vehicle overtaking via artificial potential field," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 21531–21547, Nov. 2022.
[19] H. Li, W. Liu, C. Yang, W. Wang, T. Qie, and C. Xiang, "An optimization-based path planning approach for autonomous vehicles using the dynefwa-artificial potential field," *IEEE Trans. Intell. Veh.*, vol. 7, no. 2, pp. 263–272, Jun. 2022.
[20] J. Votion and Y. Cao, "Diversity-based cooperative multivehicle path planning for risk management in costmap environments," *IEEE Trans. Ind. Electron.*, vol. 66, no. 8, pp. 6117–6127, Aug. 2019.
[21] D. H. Lee, S. S. Lee, C. K. Ahn, P. Shi, and C.-C. Lim, "Finite distribution estimation-based dynamic window approach to reliable obstacle avoidance of mobile robot," *IEEE Trans. Ind. Electron.*, vol. 68, no. 10, pp. 9998–10006, Oct. 2021.
[22] Z. Jian, S. Zhang, L. Sun, W. Zhan, N. Zheng, and M. Tomizuka, "Long-term dynamic window approach for kinodynamic local planning in static and crowd environments," *IEEE Robot. Autom. Lett.*, vol. 8, no. 6, pp. 3294–3301, Jun. 2023.
[23] O. Brock and O. Khatib, "High-speed navigation using the global dynamic window approach," in *Proc. IEEE Int. Conf. Robot. Automat.*, 1999, vol. 1, pp. 341–346.
[24] Y. Kang, D. A. de Lima, and A. C. Victorino, "Dynamic obstacles avoidance based on image-based dynamic window approach for human-vehicle interaction," in *Proc. IEEE Intell. Veh. Symp.*, 2015, pp. 77–82.
[25] S. Han, L. Wang, Y. Wang, and H. He, "A dynamically hybrid path planning for unmanned surface vehicles based on non-uniform theta* and improved dynamic windows approach," *Ocean Eng.*, vol. 257, 2022, Art no. 111655.
[26] Y. Liang, Z. Yin, L. Nie, and Y. Ba, "Shared steering control with predictive risk field enabled by digital twin," *IEEE Trans. Intell. Veh.*, vol. 8, no. 5, pp. 3256–3269, May 2023.
[27] A. A. Tilahun, T. W. Desta, A. O. Salau, and L. Negash, "Design of an adaptive fuzzy sliding mode control with neuro-fuzzy system for control of a differential drive wheeled mobile robot," *Cogent Eng.*, vol. 10, no. 2, 2023, Art no. 2276517.
[28] A. Ashagrie, A. O. Salau, and T. Weldcherkos, "Modeling and control of a 3-DoF articulated robotic manipulator using self-tuning fuzzy sliding mode controller," *Cogent Eng.*, vol. 8, no. 1, 2021, Art no. 1950105.
[29] Y. Qin, X. Wu, and J. Luo, "Data-model combined driven digital twin of life-cycle rolling bearing," *IEEE Trans. Ind. Inform.*, vol. 18, no. 3, pp. 1530–1540, Mar. 2022.
[30] Y. Wang, X. Wu, W. Zhang, and M. Wu, "Equivalent-input-disturbance-based position synchronization control of networked multiaxis motion system," *IEEE Trans. Ind. Electron.*, vol. 69, no. 8, pp. 8317–8324, Aug. 2022.
[31] H. Ji, W. Shang, and S. Cong, "Adaptive synchronization control of cable-driven parallel robots with uncertain kinematics and dynamics," *IEEE Trans. Ind. Electron.*, vol. 68, no. 9, pp. 8444–8454, Sep. 2021.
[32] Q. Xu, "Adaptive integral terminal third-order finite-time sliding-mode strategy for robust nanopositioning control," *IEEE Trans. Ind. Electron.*, vol. 68, no. 7, pp. 6161–6170, Jul. 2021.
[33] S. Arockia Samy and P. Anbalagan, "Disturbance observer-based integral sliding-mode control design for leader-following consensus of multi-agent systems and its application to car-following model," *Chaos Soliton. Fract.*, vol. 174, 2023, Art no. 113733.
[34] L. Zhao, J. Li, H. Li, and B. Liu, "Double-loop tracking control for a wheeled mobile robot with unmodeled dynamics along right angle roads," *ISA Trans.*, vol. 136, pp. 525–534, 2023.
[35] P. Ogren and N. Leonard, "A convergent dynamic window approach to obstacle avoidance," *IEEE Trans. Robot.*, vol. 21, no. 2, pp. 188–195, Apr. 2005.
[36] M. Cirillo, T. Uras, and S. Koenig, "A lattice-based approach to multi-robot motion planning for non-holonomic vehicles," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2014, pp. 232–239.
[37] S. Teng et al., "Motion planning for autonomous driving: The state of the art and future perspectives," *IEEE Trans. Intell. Veh.*, vol. 8, no. 6, pp. 3692–3711, Jun. 2023.
[38] U. Patel, N. K. S. Kumar, A. J. Sathyamoorthy, and D. Manocha, "DWA-RL: Dynamically feasible deep reinforcement learning policy for robot navigation among mobile obstacles," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2021, pp. 6057–6063.



Hongjiu Yang received the B.S. degree in mathematics and applied mathematics, the M.S. degree in applied mathematics from the Hebei University of Science and Technology, Shijiazhuang, China, in 2005 and 2008, respectively, and the Ph.D. degree in control science and engineering from the Beijing Institute of Technology, Beijing, China. From 2013 to 2018, he was an Associate Professor with the Department of Automation, Institute of Electrical Engineering, Yanshan University, Qinhuangdao, China. He is currently a Professor with the Department of Automation, School of Electrical and Information Engineering, Tianjin University, Tianjin, China. His research interests mainly include robust control/filter theory, delta operator systems, networked control systems, and active disturbance rejection control.



Ziqi Qin received the B.S. degree in control theory and control engineering from Tianjin University, Tianjin, China, in 2023. He is currently working toward the M.S. degree with the School of Electrical and Information Engineering, Tianjin University, Tianjin, China. His research interests include navigation algorithms and motion control for wheeled robots.



Yuanqing Xia (Fellow, IEEE) was born in Anhui Province, China, in 1971. He received the M.S. degree in fundamental mathematics from Anhui University, Hefei, China, in 1998, and the Ph.D. degree in control theory and control engineering from the Beijing University of AeroBnautics and Astronautics, Beijing, China, in 2001. During 2002 to 2003, he was a Postdoctoral Research Associate with the Institute of Systems Science, Academy of Mathematics and System Sciences, Chinese Academy of Sciences, Beijing, China. From 2003 to 2004, he was a Research Fellow with the National University of Singapore, Singapore, where he worked on variable structure control. From 2004 to 2006, he was with the University of Glamorgan, Pontypridd, U.K., as a Research Fellow. From 2007 to 2008, he was a Guest Professor with Innsbruck Medical University, Innsbruck, Austria. Since 2004, he has been with the School of Automation, Beijing Institute of Technology, Beijing, as an Associate Professor, and has been a Professor since 2008. His research interests include the fields of networked control systems, cloud control systems, robust control, and signal processing.



Fuyang Cheng received the B.S. degree in control theory and control engineering from Tianjin University, Tianjin, China, in 2021. He is currently working toward the M.S. degree with the School of Electrical and Information Engineering, Tianjin University, Tianjin, China. His research interests include wheeled mobile robots, digital twin, and active disturbance rejection control.

Authorized licensed use limited to: Universidad Federal de Pernambuco. Downloaded on July 31,2026 at 16:24:55 UTC from IEEE Xplore. Restrictions apply.